

Optical Microplastic Flow Tracker

Project Technical Documentation

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1. Operational Workflow

The system operates in a strict chronological sequence to ensure false positives are eliminated before optical detection occurs.

Phase 1: System Boot & Baseline Calibration

The ESP32 powers on, connects to the local Wi-Fi, and links to the Digital Twin dashboard. The **DS18B20 Temperature Sensor** reads the baseline water temperature to auto-calibrate the optical baseline, accounting for temperature-based refractive index shifts.

Phase 2: Activation & Pre-Filtration

The user triggers the "Start Flow" command via the dashboard. The ESP32 signals the 5V Relay, activating the **Peristaltic Pump**. Water is drawn and forced through an **inline mechanical mesh filter**, physically trapping large sediment, sand, and heavy mud.

Phase 3: The Turbidity Failsafe

Pre-filtered water passes through the **Analog Turbidity Sensor**. If the water is too muddy (low analog voltage), the ESP32 cuts power to the pump relay and flags an "Invalid Sample" alert. If the water is clear (high analog voltage), flow continues.

Phase 4: Core Optical Detection & Digital Twin Sync

Validated water enters the **PVC Dark Chamber**. A 650nm laser beam passes through the water to a **BPW34 Photodiode**. Microplastics scatter the laser light, causing microsecond voltage drops. The ESP32 registers these drops as detections and instantly pushes the data via Wi-Fi to update the live cloud dashboard.

2. Principle of Working

The system relies on **Optical Attenuation** combined with integrated sensor fusion to achieve edge-computed results.

- **The Optical Baseline:** Inside the dark chamber, the 650nm Laser Diode fires a continuous beam. The BPW34 Photodiode converts this light into a steady, high-voltage baseline read by the ESP32.
- **Physical Interruption (Detection):** When a microplastic particle crosses the laser path, it refracts and blocks a portion of the light. The photodiode voltage experiences a sharp, split-second drop. The ESP32 identifies this V-shaped dip as a discrete particle.
- **Failsafe Logic:** By placing the Turbidity sensor upstream of the dark chamber, the ESP32 performs intelligent sensor fusion. It refuses to count light blockages if the general water turbidity is too high, mathematically preventing the system from confusing dirt with plastic.
- **Temperature Compensation:** As fluid temperature alters water density and refraction, the DS18B20 temperature data applies a dynamic correction to the photodiode's baseline threshold, maintaining accuracy in varying environments.

3. Complete Bill of Materials (BOM)

The hardware required to construct the functional prototype.

1. Microcontroller & Power

- ESP32 NodeMCU Development Board (Built-in Wi-Fi)
- 5V Power Bank & Micro-USB Cable

2. Actuator & Control

- 5V Peristaltic Pump
- 1-Channel 5V Relay Module (with built-in optocoupler/flyback diode)

3. Sensors & Optics

- Analog Turbidity Sensor Module
- DS18B20 Waterproof Temperature Sensor
- 5V 650nm Laser Diode
- BPW34 Photodiode

4. Circuitry & Prototyping

- LM358 Op-Amp IC (For buffering the photodiode signal)
- Zero PCB (Perfboard) & Header Pins
- Resistors: 4.7k Ω (Temp pull-up), 10k Ω & 20k Ω (Turbidity voltage divider), 100k Ω (Photodiode sensitivity)

5. Plumbing & Enclosure

- 1 Meter Silicone Tubing (Sized to fit peristaltic pump nozzles)
- Inline Motorcycle Fuel Filter (Primary mechanical pre-filtration)
- Black PVC Pipe & 2 End Caps (To construct the dark chamber)
- Acrylic or MDF Boards (For the two-tier chassis)